

Experimenting with Artificial Intelligence: Programming Pathfinding Algorithms in C++ with Unreal Engine 5

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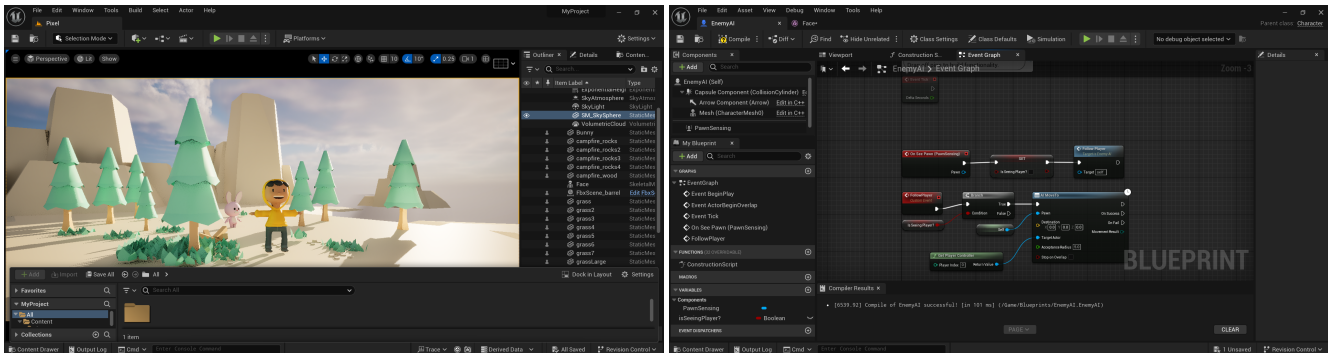


Figure 1: Students will build a tag game in Unreal Engine to learn about pathfinding algorithms for game AI. (left) Shay D. Pixel runs from the pathfinding rabbit. (right) Unreal Blueprint of the default pathfinding AI for comparison to the C++ custom algorithms students will implement.

CCS CONCEPTS

• Human-centered computing → User studies; • Applied computing → Computer games.

KEYWORDS

computer games, education

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1 INTRODUCTION

Recent years have seen a rise in machine learning applications deployed in different technological contexts. It has shaped how we experience technology and led to new doors of innovation and a mass market interest in artificial intelligence (AI). While academic research pushes the boundaries of AI through novel approaches,

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games serve as important testbeds for virtual agents. Game AI focuses on human psychology by using and looking for solutions to create fun illusions of intelligence [Roberts 2022].

In modern games, heuristic algorithms play essential roles in navigating movement, coordinating behaviors, and running animations in video games. Common examples of game AI use cases include an AI agent moving efficiently to get to a location in the world space, discovering items, and chasing a character [Group [n. d.]; Moniem 2016]. These behaviors are driven by the pathfinding process, usually with the A* Algorithm in which they are represented as graph nodes. The A* Algorithm expands upon neighboring nodes with high node metrics to find the best or most efficient path. This involves the constant calculation of the best paths between the current location and the destination.

Unreal Engine is a video game engine known for its powerful rendering technologies for detailed graphics [Zhang and Zhang 2023]. It is widely known for video game and XR development through C++ and Blueprints Visual Scripting. The built in AI heuristics use Blueprints Visual Scripting with a node-based visual programming interface. Unreal also supports C++ for more explicit programmatic control of game behavior. Unreal's AI Navigation system automates navigation meshes from scene geometries using the A* pathfinding algorithm to move objects intelligently within a scene.

2 OVERVIEW

Through collaboration and the spirit of critical making, participants will learn techniques for AI decision-making and visualize algorithm theory by creating an open-world game in Unreal Engine

5 [Epic Games 2023] (See Fig. 1). While developing our game, we take an in-depth look into graph-searching strategies through a mini-programming activity. We cover the A* heuristic found in-game AI before writing a custom algorithm. Understanding these theoretical frameworks and decision techniques is necessary to build a foundational understanding of artificial intelligence for games. This hands-on lab session focuses on studying the applications of decision-making systems and pathfinding algorithms for intelligent gameplay.

Before the workshop, the instructors prepare a UE5 project with a prebuilt game world compatible with VR. The instructors take care of all game mechanics by creating an open-world game from scratch. Packages and camera setup for non-passthrough VR experiences will be preinstalled. The project will be pre-loaded and available on the class web page.

The objective of the pathfinding game algorithms workshop is threefold. First, we provide instruction on pathfinding and heuristics. Second, we leverage AI Navigation packages to dissect the inner workings of AI pathfinding algorithms. Individually and collectively, participants step through the process of traversing and weighing graph nodes and paths. The skeleton code for their respective search algorithm then simulates real-time agent movement, beginning at the first node or the start state. Students then modify the code to compare and contrast the solutions. We finish off by building the project into a playable game. Following the class, attendees are encouraged to customize their Metahuman avatars as desired.

3 AGENDA

5 minutes: Welcome and Introductions. Welcome, overview of the course, instructor introductions, and course motivation.

10 minutes: Game Engine Scene Set-Up. An overview of the starter project's game scenes including the scripts attached, the packages included, and how to import custom assets. Participants may individually experiment with the game through instructions outlined in the game user interface.

15 minutes: Heuristics and Pathfinding Algorithms. Compare and contrast pre-determined heuristics such as A* and visualize with Unreal Blueprints and behavior trees.

15 minutes: Custom Pathfinding System. Discuss how to construct and evolve a custom heuristic in C++. Designing and implementing behavior trees

10 minutes: Building the Game. Build the project into a playable game on the platform of choice. The project accommodates AR and VR hardware.

5 minutes: Closing Remarks. Review, questions, and discussion.

4 AUDIENCE

This hands-on lab invites anyone interested in learning pathfinding algorithms for AI by creating a multiplatform video game. Our participants include a wide audience of technical and artistic practitioners of diverse backgrounds, especially those who do not work directly on games, immersive computing, or AI/ML. There will be hands-on C++ programming in Unreal Engine throughout the lab.

To get the most out of the lab, knowledge of basic code will be beneficial. Participants will apply their programming knowledge in C++.

5 BIOGRAPHIES

Deborah Yuen. is a design engineer, a professor of interactive media and game development at USC, and a UC Berkeley alumna where she received her undergraduate and graduate degrees in Human-Computer Interaction. She is a maker and advocate at heart who values creating innovative solutions that make the world a better place. Trying new things has led her to discover the world of human-computer interaction between immersive realities and the physical world of people, nature, and technology. Her research investigates the accessibility of XR-AI experiences and how these technologies drive human emotion and technology design.

Josef Spjut. is a Research Scientist at NVIDIA working on graphics and low-latency technology for competitive gaming. He is also an Adjunct Associate Professor for the Game Design, Development and Innovation Master of Engineering at Duke University. Josef is motivated by the ability to connect people to each other through the virtual worlds created by games. Previously he was a Visiting Assistant Professor in the department of Engineering at Harvey Mudd College. He received a Ph.D. from the University of Utah and a B.S. from University of California, Riverside, both in Computer Engineering.

6 CONCLUSION

By deepening the understanding of pathfinding algorithms and heuristics by programming their own, participants can develop a more informed view of AI and ML. Following the lab session, participants will be equipped with enough knowledge to begin creating indie games and expand their knowledge of data science. We believe that fostering intuition on these topics by making a game is important in bringing out creativity and imagination, contributing new perspectives to their work.

Through this lab class, students will:

- Learn how pathfinding heuristics fit in to the world of games.
- Compare and contrast A* and other pathfinding heuristics.
- Examine when hand-tuned AI behaviors are preferred.
- Build a playable game using Unreal Engine 5.
- Discover how to use games as a platform for understanding and experimenting with AI.

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